



University of Turin

Natural Sciences

Master's Degree in Computer Science

Track: Artificial Intelligence and Information Systems "Pietro Torasso"

Kolmogorov–Arnold Networks for Image Classification:

An experimental study on performance and interpretability

First Supervisor

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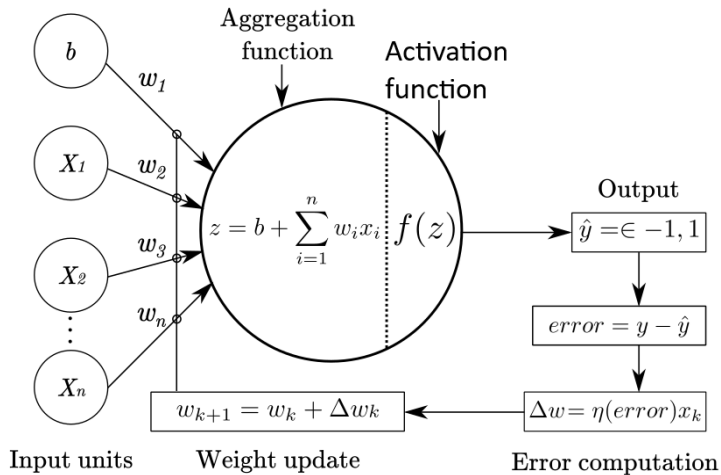
Second Supervisor

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Perceptron



KAN Neuron

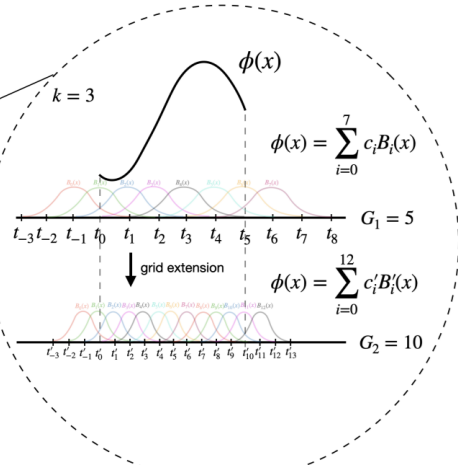
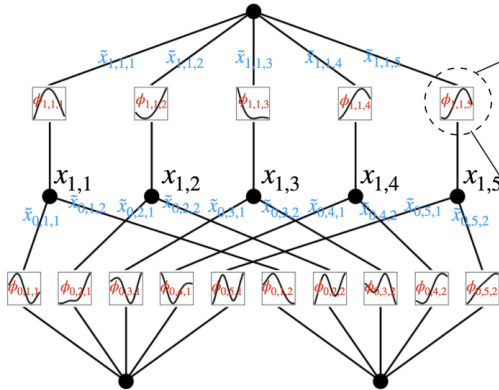


Image taken from: Ziming Liu et al., "KAN: Kolmogorov–Arnold Networks", ICLR 2025.

Differences between perceptron and KAN

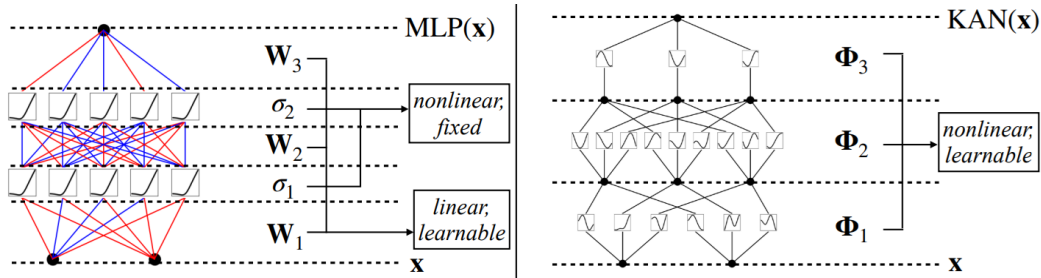


Image taken from: Ziming Liu et al., "KAN: Kolmogorov–Arnold Networks", ICLR 2025.

Hypothetical advantages:

The increase in the expressive capacity of neurons could:

- increase the abstraction of the learned function and reduce the number of neurons in the network $\rightarrow$ xAI;
- according to theoretical calculations, the network becomes more efficient as the number of parameters increases thanks to the reduction of the *curse of dimensionality* effect.

Empirical study: function approximation task with example

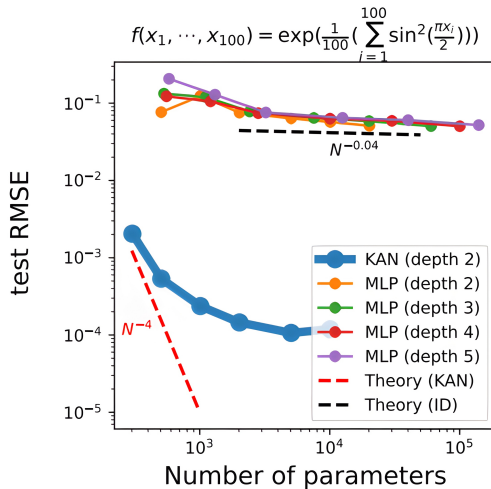
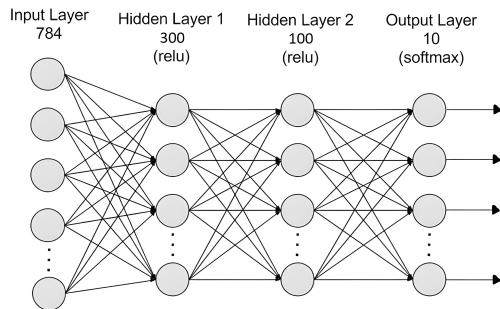


Image taken from: Ziming Liu et al., "KAN: Kolmogorov–Arnold Networks", ICLR 2025.

Why the image classification task?

- it represents a high-dimensional domain, ideal for evaluating the reduction of the *curse of dimensionality*;
- it requires high abstraction capacity, in line with the expressiveness of KANs;
- it allows a direct comparison with established architectures that are simple to replicate.

First experiment: dataset and architecture



LeNet300 architecture



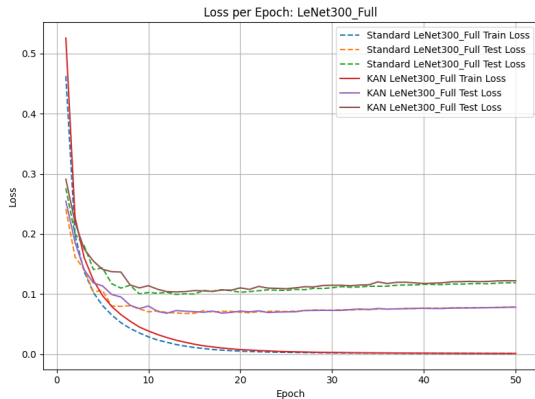
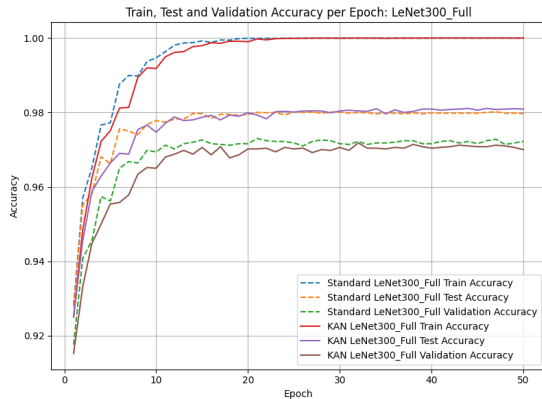
Examples from the MNIST dataset

Hypothesis

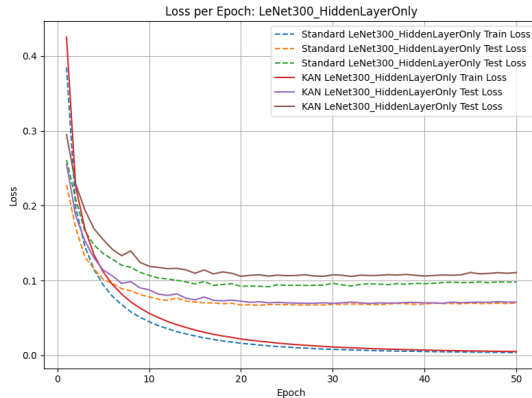
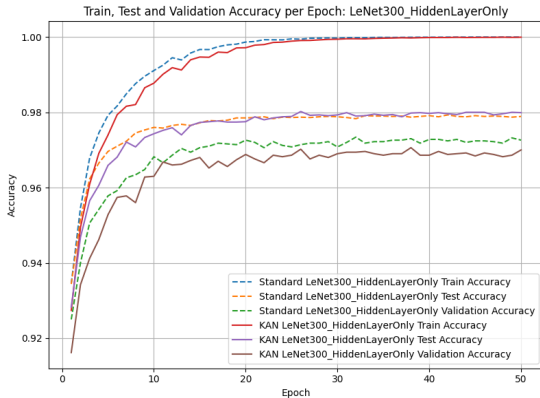
Hypothesis to be verified: the increase in expressiveness favors the adoption of KANs by reducing the number of *layers* and neurons present within the network.

Experiment: progressive ablation of the *hidden layers*.

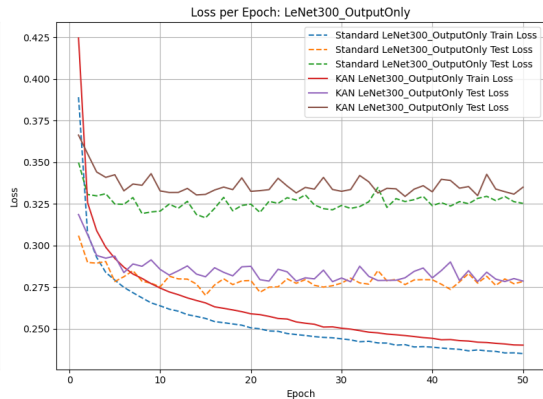
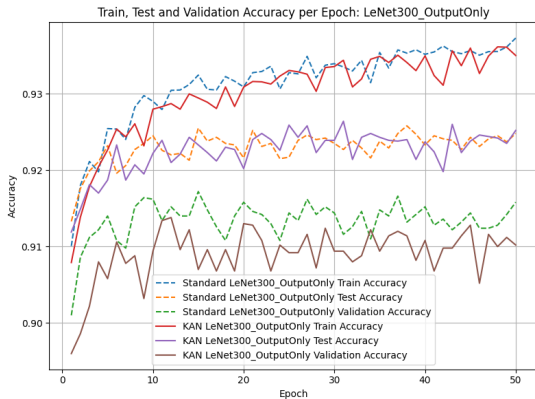
Results: all layers



Results: single hidden layer



Results: output layer only



Second experiment: dataset and architecture

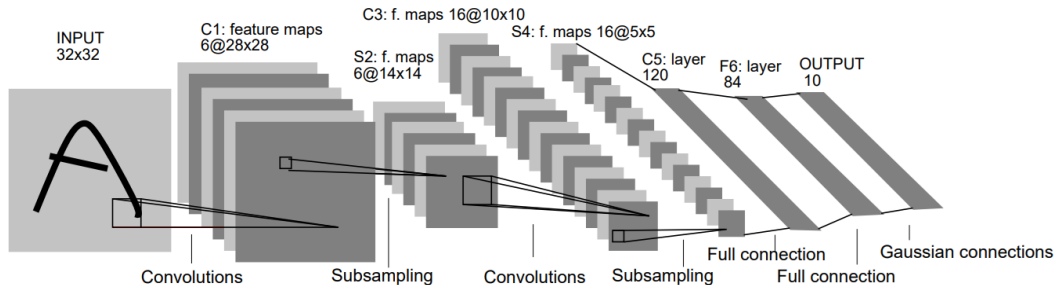
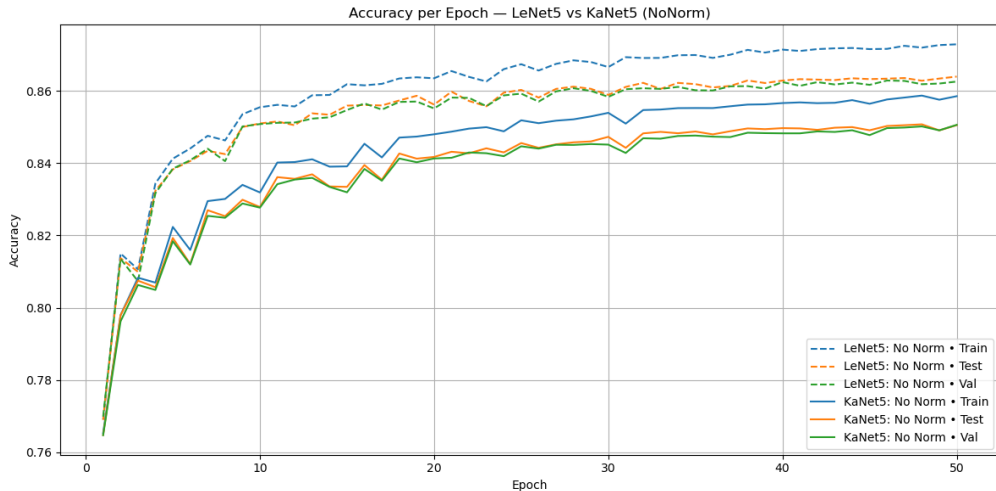
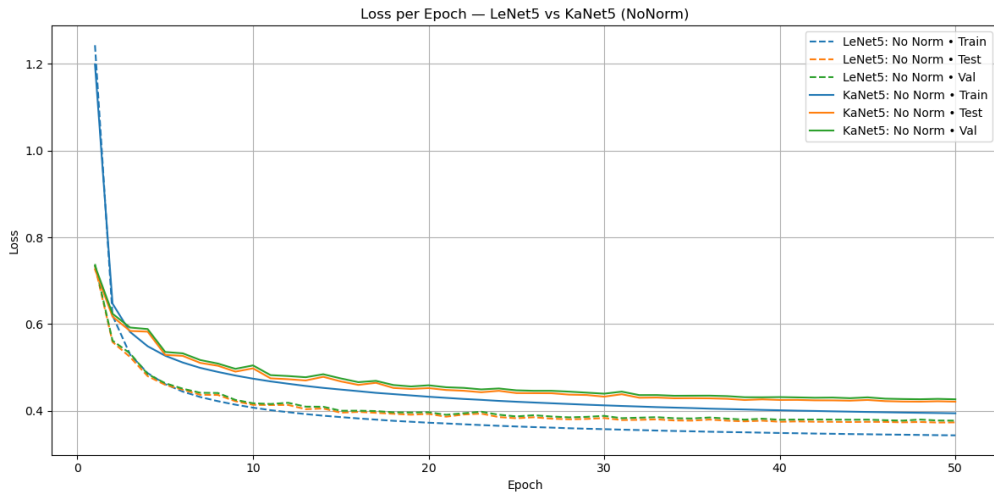


Image taken from: Yann LeCun et al., "Gradient-Based Learning Applied to Document Recognition", *Proceedings of the IEEE*, 86(11), 1998.

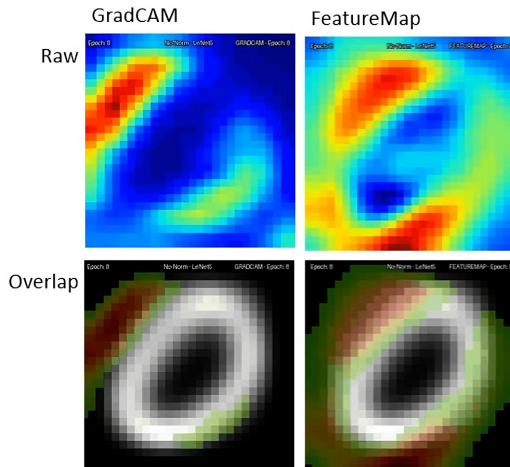
Results: Accuracy (Lenet5 vs Kanet5)



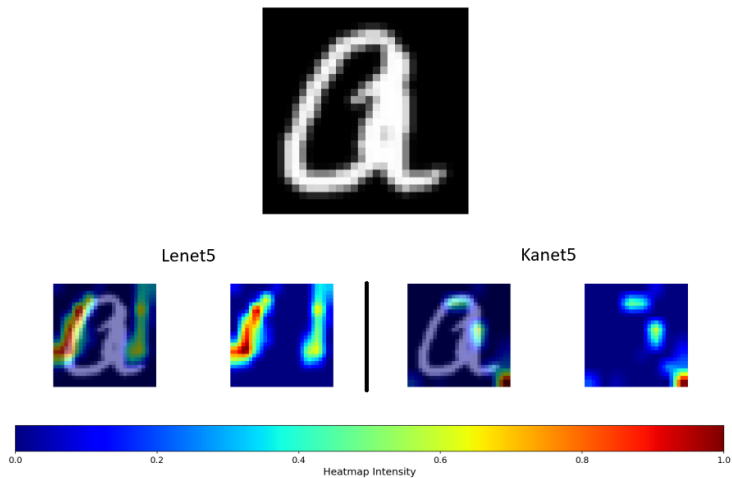
Results: Loss (Lenet5 vs Kanet5)



Explainability analysis: GradCAM



Explainability analysis: Lenet5 vs Kanet5



Explainability analysis: primitive feature detection

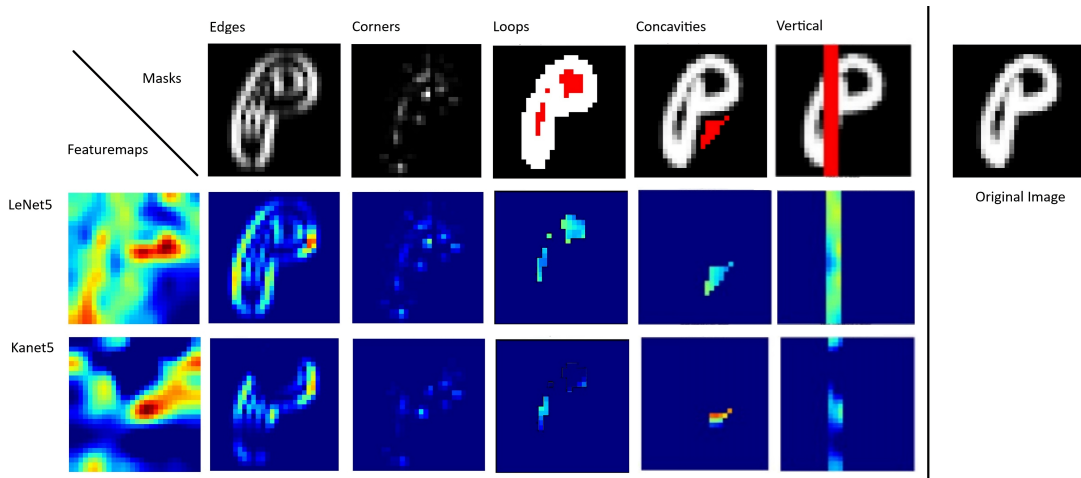
The masks used to detect primitive features were obtained through *computer vision* techniques, thus in an unsupervised manner, and include:

- Edges;
- Corners;
- Loops;
- Concavities;
- Verticality.

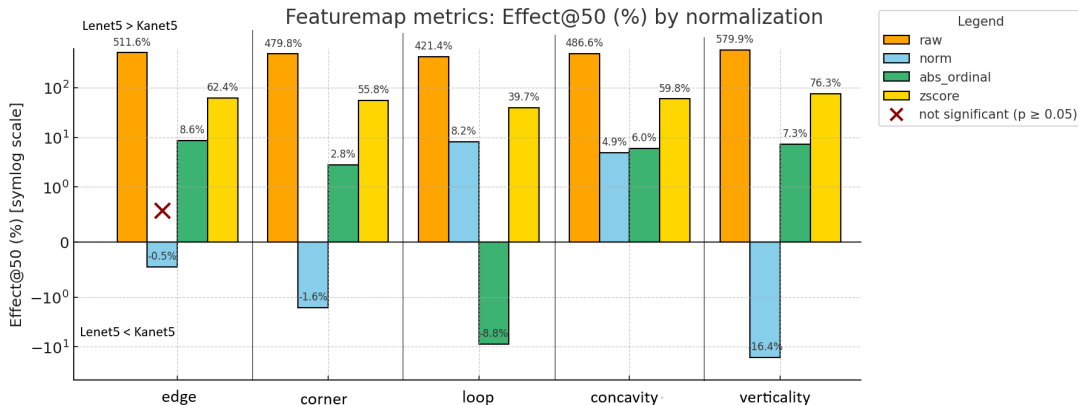
Normalizations

- **Heterogeneous distributions:** μ and σ^2 vary across layers/methods $\Rightarrow$ direct comparison is unreliable.
- **Different architectures:** LeNet5 (ReLU) $\Rightarrow$ activations ≥ 0 ; KAN $\Rightarrow$ many negative activations.
- **Cancellation effect:** means over values with opposite sign $\rightarrow 0 \Rightarrow$ hidden intensity.
- **Remedy:** normalization of gradcams and featuremaps (norm, abs_ordinal, zscore) for a common scale and comparison robust to sign/outliers.

Explainability analysis: example of feature detection

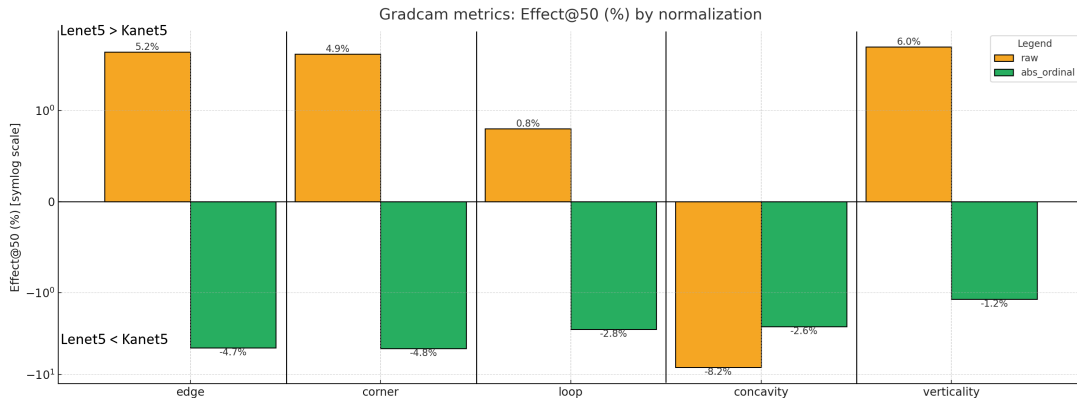


Results of the statistical analysis: FeatureMaps



Percentage increase of the median value of the mean activations, computed for the pairs (*Mask Type*, *Normalization Type*). Positive values indicate that the median of activations is higher for LeNet5 than for KaNet5, while negative values indicate the opposite.

Results of the statistical analysis: GradCAMs



The future developments of this work envisage applying KANs to:

- more complex tasks in the field of *computer vision*, such as segmentation;
- tasks that require high discriminative capacity among elements present in images;
- tasks in the presence of noise, to evaluate robustness and generalization ability.

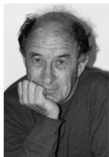
Questions?

Kolmogorov



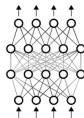
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Arnold



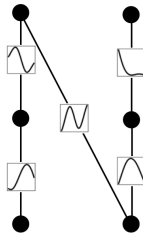
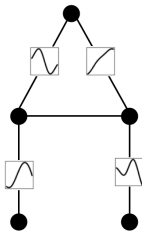
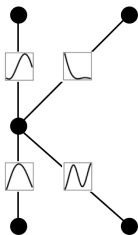
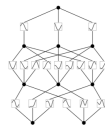
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Network



=

KAN



Mathematical

Accurate

Interpretable

Image taken from: Ziming Liu et al., "KAN: Kolmogorov–Arnold Networks", ICLR 2025.

KAN Convolution

A KAN convolution is defined as follows:

$$(\text{Image} * K)_{i,j} = \sum_{k=1}^N \sum_{l=1}^M \phi_{kl}(\mathbf{a}_{i+k,j+l})$$

Here is an example in matrix form:

$$\text{KAN Kernel} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}$$

$$\text{Image} * \text{KAN Kernel} = \begin{bmatrix} \phi_{11}(\mathbf{a}_{11}) + \phi_{12}(\mathbf{a}_{12}) + \phi_{21}(\mathbf{a}_{21}) + \phi_{22}(\mathbf{a}_{22}) & \cdots & r_{1(p-1)} \\ \phi_{11}(\mathbf{a}_{21}) + \phi_{12}(\mathbf{a}_{22}) + \phi_{21}(\mathbf{a}_{31}) + \phi_{22}(\mathbf{a}_{32}) & \cdots & r_{2(p-1)} \\ \vdots & \ddots & \vdots \\ \phi_{11}(\mathbf{a}_{m1}) + \phi_{12}(\mathbf{a}_{m2}) + \phi_{21}(\mathbf{a}_{(m+1)1}) + \phi_{22}(\mathbf{a}_{(m+1)2}) & \cdots & r_{m(p-1)} \end{bmatrix}$$

Explainability analysis: Normalizations

- **norm** (min-max on $[0, 1]$):

$$H_{\text{norm}} = \frac{H - \min(H)}{\max(H) - \min(H)}$$

- **abs_ordinal** (ranking of absolute values):

$$H_{\text{ord}}(i, j) = \text{rank}(|H_{ij}|)$$

- **zscore** (abs-z followed by min-max on $[0, 1]$):

$$A = |H|, \quad Z = \frac{A - \mu(A)}{\sigma(A)}, \quad H_z = \frac{Z - \min(Z)}{\max(Z) - \min(Z)}$$